

YIHENG LI

Foundation models over 3D medical imaging — pretraining, evaluation, training infrastructure

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6 years building AI for medical imaging · 9 peer-reviewed articles · h-index 9 · 355 citations · first author in **Cell Reports Methods**, also published in **npj Digital Medicine**

Curated 16 public and internal chest-CT collections into one re-runnable pipeline; pretrained ViT-L-class (~300M) 3D encoders on 4 GPUs with PyTorch Lightning DDP over multi-day runs; benchmarked 10 encoders on 48 downstream tasks across cohorts, scanners and sites. Before that, built self-supervised quality-control and co-registration models for a 3D MRI product in development, and fused imaging, labs and free text on a >3,000-patient multi-hospital cohort.

EXPERIENCE

Research Scientist, AI

Sep 2023 – Present

Stanford University, School of Medicine · Stanford, CA · Gevaert Lab

- **A chest-CT foundation model for lung tumor analysis** — first author, manuscript in preparation. Curated **16 public and internal CT collections** into one re-runnable harmonization pipeline the team runs. Pretrained **ViT-L-class (~300M) 3D encoders on 4 GPUs** with PyTorch Lightning DDP over multi-day runs, with mixed precision, gradient checkpointing and checkpoint-resume across preemption; profiled and cleared dataloader, storage-I/O and memory bottlenecks. Benchmarked **10 encoders on 48 downstream tasks in 5 families** (detection/segmentation, radiological, mutational, prognostic, generative) under patient-level splits, repeated seeds, ablations, and Wilcoxon tests with multiple-testing correction.
- **Benchmarking Classification Tasks of Chest X-ray AI Across Multinational Public and Private Datasets: a retrospective diagnostic accuracy study** — co-first author; submitted to *The Lancet Digital Health*, preprint arXiv 2505.16027, 2025. Evaluated **8 models** (5 vision-language foundation models, 3 CNN baselines) on **36 tasks across 9 multinational datasets**, adult and pediatric, measuring zero-shot accuracy and robustness under demographic shift.
- **A 3D lung lesion variational autoencoder** — first author, *Cell Reports Methods* 4(2), 2024. Trained a 3D β -VAE on **unlabeled** CT volumes (SSIM 0.774, PSNR 26.1) and synthesized lesions of controlled size by latent manipulation. Transferred the embeddings to nodal stage and KRAS mutation prediction, **matching fully supervised models**.

Deep Learning Research Scientist

Aug 2021 – Aug 2023

Subtle Medical Inc. · Menlo Park, CA

- **Self-supervised quality control and co-registration for multi-contrast 3D MRI** — ISMRM 2022 and 2023, RSNA 2022. Built custom augmentation plus Jacobian and cycle-consistency losses to suppress unrealistic deformation fields; improved **SSIM ~40% and PSNR ~50% over baseline** on BraTS and lumbar-spine data, for a new imaging product in development.
- **Self-supervised coverage detection on abdominal CT and head MRI**. Adapted keypoint detection with distance-weighted losses and masking and ran a systematic loss search, for **~30% improvement** on proprietary clinical data. Scoped the problem and mentored the intern who carried it.

Research Assistant

Jul 2020 – Jul 2021

Stanford University · Stanford, CA · Camarillo, Chaudhari (AIMI) and Gevaert Labs

- **AI-based analysis of CT images for rapid triage of COVID-19 patients** — *npj Digital Medicine* 4:75, 2021 (53 citations). Fused **9,943** imaging-derived features with lab values and free text; trained $n=1,662$, externally validated $n=1,362$ across hospitals; **AUROC 0.85–0.94**.
- **Head-impact subtyping and brain-strain prediction** — 6 papers, **200+ citations**. Clustered high-frequency head kinematics by spectral density and modeled injury outcomes; built a U-Net vertebral segmentation pipeline for Stanford AIMI's Opportunistic CT.

SKILLS

Languages and frameworks: Python, C++, CUDA, SQL, Bash, R · PyTorch, PyTorch Lightning, HuggingFace Transformers, MONAI, TensorFlow, scikit-learn, NumPy, pandas, OpenCV · DICOM, NIFTI

Methods: foundation models, transformers, self-supervised and contrastive learning (SimCLR, BYOL, DINO), generative models (VAE, diffusion), vision-language models (CLIP), multimodal fusion, machine learning under distribution shift, 3D registration and segmentation, multi-instance learning, survival analysis

Training and infrastructure: multi-GPU data-parallel training (Lightning DDP), mixed precision (AMP / bf16), gradient checkpointing and accumulation, checkpoint-resume under preemption, SLURM job scheduling, throughput profiling · Git, Docker, MLflow, Weights & Biases, unit testing, Python packaging, AWS, GCP, Azure

EDUCATION

Stanford University — M.Sc. in Biomedical Informatics

Sep 2019 – Jun 2021

Shanghai Jiao Tong University — B.Sc. in Resource and Environmental Science

Sep 2015 – Jun 2019

PUBLICATIONS

9 peer-reviewed journal articles · h-index 9 · 355 citations · **first or co-first author on 5**, including *Cell Reports Methods* 2024, *J. R. Soc. Interface* 2021 (83 citations) and *Ann. Biomed. Eng.* 2021 (42 citations). Guest lecturer, Stanford BMDS 221. Peer reviewer for 9 journals, 20+ manuscripts. Full list on Google Scholar.